

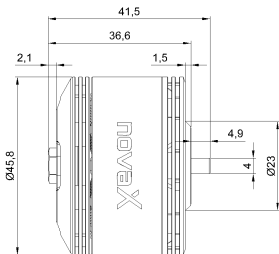
AM-4116

User Guide

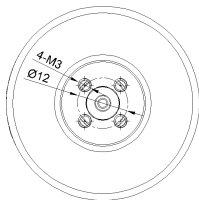


AM-4116 Dimensions

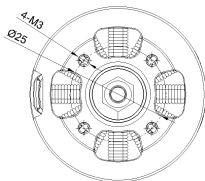
• Dimensions (mm)



Side



Bottom · mount



Top · propeller side

AM-4116 Specifications

• Basic Specifications

Stator Dimensions	$\Phi 45.7 \times 43$ mm
Slots / Poles	12N14P (7 pole pairs)
Motor Weight	210 g
KV Value	KV450
Rated Voltage	22.2 V (6S LiPo)
Max Current	48.8 A
Max Power	1154 W
Recommended Prop	15 ~ 16.5 inch (15×5.0 / 15×8.0 / 16×5.4)
Operating Temp	-20 ~ +85 °C

AM-4116 Performance Data

6S LiPo, sea-level static thrust stand. Measured values; rated specs may differ.

• Propeller 15.5 × 5.0 • BLDC ESC

Throttle	Thrust (g)	Current (A)	Power (W)	Eff. (g/W)	Speed (rpm)
40%	830	3.8	96	8.61	4146
50%	1247	6.7	168	7.42	5079
60%	1781	10.8	270	6.60	5994
70%	2314	16.2	402	5.75	6800
80%	2893	22.4	555	5.22	7579
90%	3507	30.1	742	4.73	8310
100%	4069	36	882	4.51	8767

AM-4116 Performance Data

• Propeller 16.5 × 5.4 • BLDC ESC

Throttle	Thrust (g)	Current (A)	Power (W)	Eff. (g/W)	Speed (rpm)
40%	976	4.6	116	8.44	4095
50%	1554	7.9	199	7.82	5003
60%	2085	13	324	6.43	5875
70%	2631	18.4	458	5.75	6652
80%	3294	25.9	639	5.16	7401
90%	3974	34.9	856	4.64	8076
100%	4530	41.1	1005	4.43	8522

• Propeller 15.5 × 5.0 • FOC ESC (Combo)

Throttle	Thrust (g)	Current (A)	Power (W)	Eff. (g/W)	Speed (rpm)
40%	607	2.6	66	9.16	3544
50%	963	4.7	120	8.05	4425
60%	1372	7.7	195	7.05	5311
70%	2141	13.9	349	6.13	6240
80%	2842	22.2	554	5.13	7139
90%	3638	32.3	800	4.55	—
100%	4455	42.7	1051	4.15	9001

AM-4116 Performance Data

• Propeller 16.5 × 5.4 • FOC ESC (Combo)

Throttle	Thrust (g)	Current (A)	Power (W)	Eff. (g/W)	Speed (rpm)
40%	739	3.1	78	9.43	3524
50%	1180	6	150	7.87	4425
60%	1648	9.5	239	6.89	5311
70%	2259	14.4	360	6.28	6182
80%	3019	22.3	556	5.43	7095
90%	3985	32.6	807	4.94	8016
100%	4928	47	1154	4.19	8835

AM-4116 Installation



1. **Mounting** — Fix the motor to the arm through the bottom mounting holes. Follow the screw size and spacing on the drawing and use screws that do not reach the windings.
2. **ESC wiring** — Connect the three motor wires to the ESC outputs. To reverse the rotation, swap any two wires or reverse it in the ESC settings.
3. **Spin check** — Without a propeller, run at low throttle and check the direction and for abnormal noise.

AM-4116 Installation

4. Propeller — Fit a propeller matching the rotation and tighten the nut firmly.
Recommended: 15 ~ 16.5 inch (15×5.0 / 15×8.0 / 16×5.4).

AM-4116 Safety Notes

- Do not exceed the rated voltage. Rated: 22.2 V (6S LiPo).
- Use an ESC rated with margin above the motor maximum current. Max current: 48.8 A.
- Always remove the propeller while wiring, configuring or bench testing.
- The motor can be hot right after operation; let it cool before touching.
- Keep metal debris away from the magnets; stop using the motor if the shaft is bent or the bearings are noisy.



<Detailed Manual>

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